

Guidance and surveillance capability for bystander *Staphylococcus aureus* tetracycline resistance under doxy-PEP: a three-stream evidence audit

A. C. DEMIDONT*

Nyx Dynamics, LLC, Fairfield, Connecticut, USA

Running title: Surveillance and doxy-PEP bystander *S. aureus* resistance

*Corresponding author. Nyx Dynamics, LLC, Fairfield, Connecticut, USA. Tel: +1-215-901-2366. E-mail: acdemidont@nyxdynamics.org

ORCID: 0000-0002-9216-8569

Synopsis

Background. Doxycycline post-exposure prophylaxis (doxy-PEP) prevents bacterial sexually transmitted infections and has scaled into guidance internationally. Doxycycline also selects tetracycline resistance beyond its target, and *Staphylococcus aureus*—a commensal and pathogen for which doxycycline is one of a limited number of oral options—is the natural bystander.

Objectives. To determine whether the guidance and surveillance deployed around doxy-PEP could follow the bystander-resistance signal the pivotal trial's final analysis resolved, rather than to estimate a population effect.

Methods. A reproducible, public-data audit of three evidence streams—clinical trials, governmental guidance and deployed US surveillance—each scored against one detectability framework: the within-exposed relative risk it could detect, benchmarked to an empirical colonisation estimate.

Results. The final randomised DoxyPEP analysis reported a 3.89-fold higher hazard of incident doxycycline-resistant *S. aureus* under doxy-PEP (95% CI 1.42–10.68; 68 of 393 versus 5 of 163 incident events), with no effect on colonisation clearance. Yet 0 of 6 outbreak-matched

governmental guidelines (0 of 10 coded) required *S. aureus* monitoring, and 0 of 12 established US surveillance systems linked doxy-PEP exposure to an *S. aureus* tetracycline phenotype at a common population denominator. Under realistic exposure density and isolate volumes, an ordinary population system would need an implausible within-exposed effect ($RR \approx 6\text{--}28$) to move a rate.

Conclusions. The upstream trial generated information the downstream architecture is not built to receive. Following the signal requires sentinel surveillance linking doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator. No population-level causal claim is made.

Introduction

Doxycycline post-exposure prophylaxis (doxy-PEP) substantially reduces bacterial sexually transmitted infections (STIs) and has moved quickly from trial result into guidance at every level, from the World Health Organization through national and municipal health departments.^{1,2} Every dose is selective pressure, and not only on the organisms it targets. *Staphylococcus aureus* is the consequential bystander: it colonises the anatomical sites these trials swabbed and carries mobile tetracycline resistance determinants.³ For two decades, community MRSA in these networks has been documented not as steady-state carriage but as clustered, outbreak-pattern transmission, with the USA300 lineage moving through sexual networks in San Francisco, Tokyo and Chicago.^{4,5} The organism colonising these networks is the same one carried on skin community-wide, so resistance selected within the network could, if transmitted, extend beyond it. Tetracycline-resistant USA300 was in fact documented in an MSM-predominant community health population two decades ago, although by the doxycycline-sparing *tet(K)* mechanism, with those isolates remaining doxycycline-susceptible.⁶ Doxycycline is one of a limited number of oral options for outpatient skin and soft-tissue infection, a choice often made empirically before any methicillin result returns,^{7,8} so tetracycline susceptibility can matter clinically before methicillin status is known.

Clinicians who prescribe doxy-PEP have registered antimicrobial-resistance concerns di-

rectly, but those concerns have largely been framed around familiar targets. In a survey of Italian STI-clinic dermatologists, 91.7% named methicillin-resistant *S. aureus* (MRSA) selection as a worry, tied with gonococcal resistance as the highest-rated concern,⁹ while a systematic review of 1840 healthcare professionals across four countries likewise placed antimicrobial resistance among the leading concerns about doxy-PEP.¹⁰ That framing does not capture the full tetracycline-resistance question in *S. aureus*. In a military cohort of 168 personnel receiving doxycycline for malaria prophylaxis, overall tetracycline-class resistance in *S. aureus* did not increase, but *tet(M)* determinants were significantly more common among doxycycline-exposed personnel ($P = 0.031$), showing that aggregate susceptibility can remain stable while the underlying resistance mechanism shifts.¹¹

The trial literature has now answered the tetracycline question directly. In the final DoxyPEP analysis, among participants free of doxycycline-resistant *S. aureus* at baseline, doxy-PEP carried a 3.89-fold higher hazard of incident doxycycline-resistant *S. aureus* (95% CI 1.42–10.68; 68/393 versus 5/163; $P = 0.0044$), while the hazard of clearing colonisation did not differ between arms (HR 1.01, 0.69–1.46).¹² The trial’s authors called for public-health surveillance of antimicrobial resistance in *S. aureus* and other commensals. One national programme has since begun to address that call: the UK Health Security Agency’s 2026 doxy-PEP monitoring plan for England proposes tetracycline-resistance surveillance in bystander pathogens, naming *S. aureus*.¹³

That call defines the present question. We ask not whether doxy-PEP drives population-level resistance—a claim we do not make—but whether the guidance and surveillance deployed around doxy-PEP could follow the signal the trial resolved, at the population level where the clinical stake lies. The clinically decisive axis is tetracycline (doxycycline) susceptibility used in empirical oral therapy, not the methicillin axis around which staphylococcal surveillance is organised. We audit the three places an answer would have to come from—the trials, the guidance and the deployed surveillance—each against a single detectability framework.

Materials and methods

45

Ethics

46

This study analysed publicly available guidance, surveillance documentation and aggregate published data; it did not involve individual-level human-participant data, and institutional review board approval was therefore not required.

47

48

49

Design

50

The study is a documentary and quasi-computational audit assembled in three streams, collected and analysed in sequence: clinical trials (Stream A), the governmental guidance built on them (Stream B) and the population surveillance meant to monitor them (Stream C). No individual-level or restricted data were used; every input is public. Sources were located, read, coded with page/section/table locators and SHA-256-pinned before analysis; analysis decisions that could go either way were date-logged before results were known; and every figure and table regenerates from a single command over a test-gated pipeline archived with a citable DOI.¹⁴ Large language model use is disclosed in the Acknowledgements.

51

52

53

54

55

56

57

58

Detectability framework

59

The three streams were evaluated within a common detectability framework: whether, and at what effect magnitude, each measurement architecture could resolve a bystander-resistance signal. For the trial and surveillance streams, detectability can be expressed quantitatively against an externally specified relative-risk benchmark; for guidance, the corresponding question is whether *S. aureus* measurement is specified at all. The empirical benchmark is Soge *et al.*¹⁵ (tetracycline-resistant *S. aureus* colonisation, 18% versus 8% in doxy-PEP users with >3 doses/month versus non-users; RR $\approx$ 2.25); a cross-organism gonococcal RR $\approx$ 1.42 is retained as a prespecified, more conservative secondary benchmark.

60

61

62

63

64

65

66

67

Hazard ratio and risk ratio are different estimands, so the trial estimate is used only as a magnitude reference, not as a numerically equivalent effect measure.

68

69

Sources and coding

The author identified, retrieved and screened all source documents for eligibility; each eligible document was then coded from full text against a prespecified schema, every value carrying a page, section or table locator. For the guidance corpus (Stream B), a first LLM pass coded all 10 documents against the schema. A second, blinded pass—run through a chat interface rather than the command-line agent, but within the same model family—separately re-coded a 5-document subsample (25 field-codings) for reliability. Disagreements were resolved by the author against the source. Per-field agreement is reported as raw concordance, which measures the reproducibility of the coding protocol rather than its accuracy; because the passes share a model family their errors are not independent, so a Cohen’s κ is given for context only (Limitations). The surveillance enumeration (Stream C) was coded by a single pass; its inclusion and exclusion list, with a stated reason per system, is given in the Supplementary data and reflects systems as deployed as of August 2026. All coding prompts and outputs are in the public repository.

Stream A—trials

For each primary-trial *S. aureus* arm comparison (first-party published data only), we computed, from arm sizes and observed control rate, the exact (Fisher) power to detect the externally fixed benchmark effect and the minimum detectable RR at 80% power; fixing the effect externally makes this a design-based sensitivity analysis rather than observed power. The *S. aureus* assay is doxycycline throughout (E-test, MIC $\geq$ 16 mg/L). We tabulated how the resistance result was reported across trial venues and secondary syntheses (numerator, denominator basis, conditioning set and terminology). The final trial performed a participant-level time-to-event analysis itself; we cite that randomised result directly and did not re-estimate it. Cross-sectional carriage alone does not resolve the clustered transmission structure of *S. aureus* in these networks; the supporting dispersion, trend-test degradation and outbreak-shape analyses are MRSA-scoped (MRSA being a subset, not the analytic unit) and are reported in the Supplementary data (available as Supplementary data at JAC Online).

Stream B—guidance

97

The denominator was anchored to the same clustering evidence used in Stream A: US jurisdictions with documented community-associated MRSA/USA300 transmission in men who have sex with men.^{4,5} We took governmental public-health guidance only (clinic and provider-organisation documents excluded, with reasons). Each document was coded from full text, one record per document, every finding carrying a locator, for whether it requires, suggests or omits *S. aureus* monitoring and whether it names, counsels on or measures the organism (coding provenance above). An independence check separated genuinely independent declines from documents that defer to, or are adapted from, one another.

98
99
100
101
102
103
104
105

Stream C—surveillance

106

We examined a prespecified set of 12 established US surveillance and data systems (federal, state or metropolitan) plausibly capable of capturing, at the required level, either doxy-PEP/doxycycline exposure or *S. aureus* AMR phenotype, and coded each on three capabilities: exposure-side capture, phenotype-side capture (of the *S. aureus* tetracycline unit, not the MRSA subset) and the join, recording the geographic resolution of each side. A supporting analysis asked whether, even at a common denominator, the exposed subgroup is dense enough to move a population rate. The exposed population is indexed on the combined PrEP and people-living-with-HIV male count, the primary specification, because the trial enrolled both (AIDS Vu 2024 state PrEP and prevalence). The dilution fraction is $f = ((\text{male-PrEP} + \text{male-PLWH}) \text{ density} / 10^5) \text{ uptake } d \kappa$, where d is the share of users taking >3 doses/month (the subgroup carrying the benchmark effect) and κ the isolate-enrichment factor. Because f tracks exposure density rather than headcount it is population-independent, and the minimum detectable change in a proportion near baseline R_0 inverts to $RR_{\text{needed}} = 1 + \text{MDE} / (f R_0)$. Parameter choices were deliberately mixed: isolate volume was set high to favour detection, while the realistic under-sampling ($\kappa = 0.5$) and dose-threshold ($d = 0.5$) factors bound against it. Both ranges and the resulting RR_{needed} are reported; full grids and the metropolitan break-even are in the Supplementary data.

107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123

Scope

124

The corpus and the surveillance enumeration were restricted by design: Stream C is a prespec- 125
ified set of US systems, and Stream B is US governmental guidance with the World Health 126
Organization included only as the international apex reference. The resulting counts describe 127
US architecture, not a global census. 128

Results

129

Stream A: trial measurement and reporting of the *S. aureus* endpoint

130

The final DoxyPEP analysis ran the participant-level test the interim had not—time to first 131
detection of doxycycline-resistant *S. aureus* by arm, standard-of-care censored at crossover, by 132
Cox proportional hazards—and returned HR 3.89 (95% CI 1.42–10.68; $P = 0.0044$), with 133
colonisation clearance unchanged (HR 1.01).¹² The 2023 interim, by contrast, was presented 134
as reassuring on a cross-sectional resistance fraction and was underpowered for any externally 135
specified effect (0 of 4 comparisons powered; minimum detectable RR 3.8–5.1 against the 136
benchmark 1.42–2.25). The hazard ratio is not comparable to that cross-sectional minimum: 137
the estimand, not the sample size alone, became adequate to the question. 138

Two features of the interim reporting remained relevant to downstream interpretation. First, 139
a unit mismatch: the trial’s data table is headed “phenotypic resistance to the tetracycline an- 140
tibiotic class” while its methods define the *S. aureus* assay as doxycycline, and the federal 141
guideline then describes the doxycycline result as “tetracycline resistance in *S. aureus*”.² Be- 142
cause efflux and ribosomal-protection mechanisms differ in whether doxycycline still works 143
(below), an endpoint scored as “the tetracycline class” does not answer the treatment-relevant 144
question. Second, the signal was reported over differing, incompletely documented denomina- 145
tors and then propagated into secondary reporting under further contrasts (Table 1): the resistant 146
numerator changes from 5 to 16 to 28 and the denominator basis shifts from colonised carriers 147
to all-swabbed. The percentage therefore propagated more consistently than the denominator, 148
contrast and terminology needed to interpret it. The individual-level data that could reconcile 149

Table 1: Changes in reporting of the DoxyPEP *S. aureus* resistance signal across publication and guidance venues. Numerators, denominators, conditioning sets, and contrasts differed across reports. The table is intended to show how the reported percentage was propagated while the underlying denominator and estimand changed; rows should not be interpreted as equivalent effect estimates.

Venue	Reported result	Frame / conditioning set	What travels downstream
NEJM appendix ¹	5/31 = 16%	Month-12 doxy-PEP arm; resistant among carriers	Numerator and carrier denominator explicit
CROI abstract ¹⁷	16/137 = 12%	Doxy-PEP arm; all-swabbed denominator	Different numerator/denominator basis
CDC guideline ²	28/222 = 13%; summarised 5%→13%	Doxy-PEP arm; guideline frame	Percentage retained, provenance less transparent
JAMA Insights ¹⁸	5% versus 4%	Doxy-PEP versus control; “infected with”	Between-arm percentage only; counts/timepoint dropped
JAMA Synopsis ¹⁹	5%→16%	Within arm, among <i>S. aureus</i> -isolated	Carrier-conditioned within-arm percentage (baseline to month 12)
Lancet ID final ¹²	HR 3.89 (1.42–10.68); 68 of 393 vs 5 of 163	Participant-level time-to-event; baseline-negative, as-randomised	Not cited by any coded guidance, including two revised after its March 2025 online publication

these first-party reports are not publicly available.¹

Across trials the phenotype was measured on incompatible terms: DoxyPEP measured doxy-cycline resistance by E-test over two denominators; DuDHS measured it by disc diffusion in single-digit carriers; DOXYVAC measured only MRSA carriage—a methicillin, not a tetracycline, quantity, and one that itself rose within the doxy-PEP arm (1.8% to 9.9%) while being reported as a between-arms null.¹⁶ A pooled estimate is not defensible. Even the intensively surveilled gonococcus was phenotyped in fewer than one diagnosis in five (resistance testing unavailable for 83%, 212/256).¹⁷

The estimand shift did not propagate downstream: no guidance document in the corpus cites the final time-to-event analysis or any hazard ratio, and the two documents revised after its online publication in March 2025—WHO (2026) and San Francisco (May 2026)—retained a cross-sectional or generic frame. Non-citation is telling here because this corpus demonstrably transmits numbers when it has them: CDC carried the cross-sectional 5%→13% figures and WHO cites Soge’s colonisation estimate.¹⁵

Stream B: *S. aureus* monitoring requirements in governmental guidance

The governmental guidance shows a consistent within-document asymmetry: the efficacy question received a graded, systematic evidence review, while the AMR-harm question—explicitly including resistant-pathogen development—received an ungraded review in the same document. Of the six documented-outbreak jurisdictions with governmental doxy-PEP guidance (San Francisco, Chicago, New York City, Boston/Massachusetts, Los Angeles County and San Diego), 0 require or suggest *S. aureus* monitoring. The decline is not an artefact of copying: four jurisdictions are independently authored, one (Los Angeles County) is adapted from San Francisco and one (Massachusetts) from an external template. Nor is it for want of naming the organism—three of the six name *S. aureus* specifically, two invoke only a generic “microbiome” risk and one is silent; even where named, none measure it. Across the full coded corpus (10 documents: CDC, WHO and eight US jurisdictions) the result is identical: none require *S. aureus* monitoring. Double-coding of a 5-of-10-document subsample (25 field-codings) gave 92% raw agreement (23/25); the one field with meaningful disagreement (bystander treatment; $\kappa = 0.58$) is reported unrounded. Providers are nonetheless required to counsel patients about resistance in commensal organisms—a harm no system measures.

Stream C: surveillance capability and exposure–phenotype linkage

Following the signal requires one system to carry, at a common population denominator, both who is exposed to doxy-PEP and what tetracycline phenotype *S. aureus* carries. Across this prespecified set of twelve systems, none does (Figure 1). Two systems capture exposure at a population denominator (AIDSVu and CDC AtlasPlus, as a pre-exposure prophylaxis proxy, not doxy-PEP itself),²⁰ and zero capture *S. aureus* tetracycline resistance at a population denominator, so population-level linkage is not possible.

The absence is structural, not incidental. Wherever *S. aureus* is surveilled at a population denominator the reported axis is methicillin: the one population-based platform, Active Bacterial Core surveillance/Emerging Infections Program (ABCs/EIP), reports invasive MRSA; the National Healthcare Safety Network reports MRSA events; and the only routine source of *S. aureus* tetracycline susceptibility, the facility cumulative antibiogram, is non-standardised

and carries no population denominator, because CLSI M39 mandates the MRSA/MSSA split and makes tetracycline a selectively suppressible supplemental agent. The National Antimicrobial Resistance Monitoring System excludes *S. aureus* altogether, and tetracycline-resistant *S. aureus* is not nationally notifiable in the United States. A system could be built—most directly by adding *S. aureus* tetracycline susceptibility to ABCs/EIP, which already holds the denominator—but none is deployed.

Nor could an ordinary population system recover the missing side by collecting more isolates. Indexing the exposed population on the combined PrEP and people-living-with-HIV male count—the primary specification, because the trial enrolled both (Materials and methods)—the realistic-cell within-exposed effect required to move a population rate runs to $RR \approx 6\text{--}28$ across the design-effect range (Figure 2); the more restrictive PrEP-only indexing gives $RR \approx 14\text{--}64$, which overstates infeasibility. Either way it is far above any plausible effect. Detection appears in a single grid cell that maxes every generous assumption at once (RR_{needed} 1.03, or 1.79 under single-comparison inference; parameters in the Supplementary data). Detectability turns on quantities no deployed system reports—exposure density, isolate enrichment and dosing frequency. Against the more conservative cross-organism benchmark (RR 1.42), the best metropolitan cell needs about twice the combined male exposure density of the densest US geography (Washington, D.C.), and realistic cells far more. The only configuration that restores sensitivity is to sample *S. aureus* directly from doxy-PEP-exposed persons, converting surveillance into a sentinel cohort.

Alt text: A schematic summarising the 12-system surveillance audit: exposure is captured only as a PrEP proxy and the *S. aureus* tetracycline phenotype at a population denominator nowhere, so no system links exposure to phenotype.

Alt text: A logarithmic relative-risk axis with a shaded benchmark band at 1.42 to 2.25; the trial hazard ratio 3.89 sits above the band, surveillance detectability thresholds of about 6 to 28 (primary combined-denominator specification; 14 to 64 under the more restrictive PrEP-only specification) sit far to the right, and guidance has no threshold.

The three streams are summarised against the detectability framework in Table 2; a PubMed literature-coverage analysis of the bystander organism (361 papers, of which few name or mea-

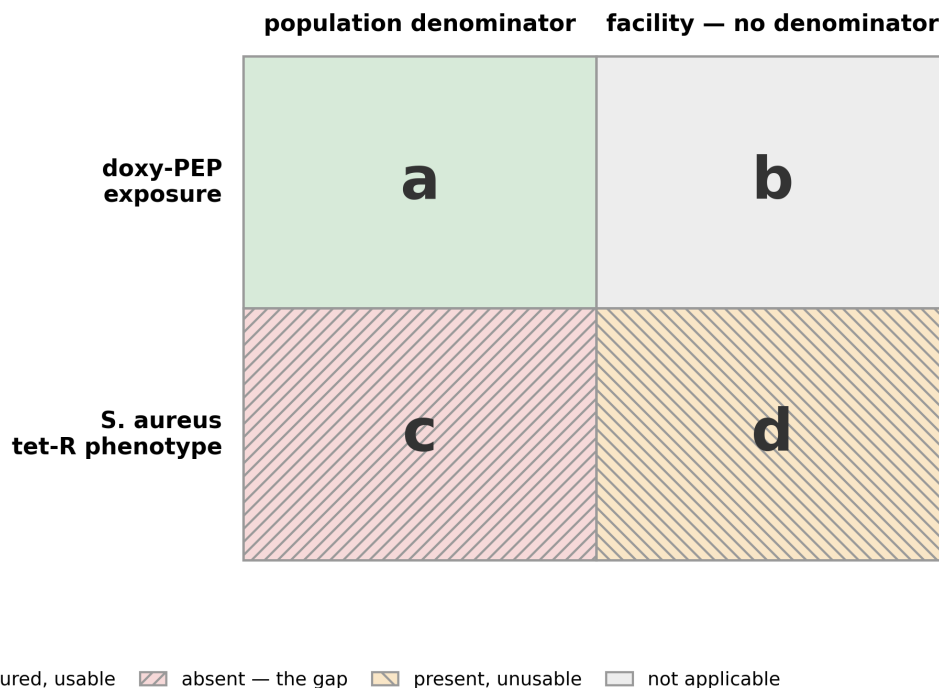


Figure 1: Surveillance architecture for doxy-PEP exposure and *Staphylococcus aureus* tetracycline resistance. Schematic summary of the 12 US surveillance and data systems assessed. Each quadrant represents the intersection of the measurement target (rows) and denominator structure (columns): (a) doxy-PEP exposure at a population denominator—captured, although only through PrEP-based proxies (AIDSVu and CDC AtlasPlus); (b) doxy-PEP exposure in facility-level data without a population denominator—not applicable to the population-level linkage question; (c) *S. aureus* tetracycline phenotype at a population denominator—absent, constituting the principal surveillance gap; and (d) *S. aureus* tetracycline phenotype in facility-level data—present in cumulative antibiograms but unusable for population linkage because no population denominator is retained. No assessed system therefore occupies both (a) and (c) within a common linkage structure, and 0 of 12 systems link doxy-PEP exposure to *S. aureus* tetracycline phenotype at a common population denominator.

sure it) is reported in the Supplementary data (S5).

221

Discussion

222

The pivotal trial has now resolved the bystander question it was long thought unable to reach: 223
 a significant increase in incident doxycycline-resistant *S. aureus* under randomised exposure, 224
 with no matching effect on carriage, and a call from its authors for surveillance of exactly this.¹² 225
 Across the two systems that would have to receive it—guidance and deployed surveillance— 226
 they cannot: guidance instruments nothing (0/6; 0/10) and no deployed system joins exposure 227

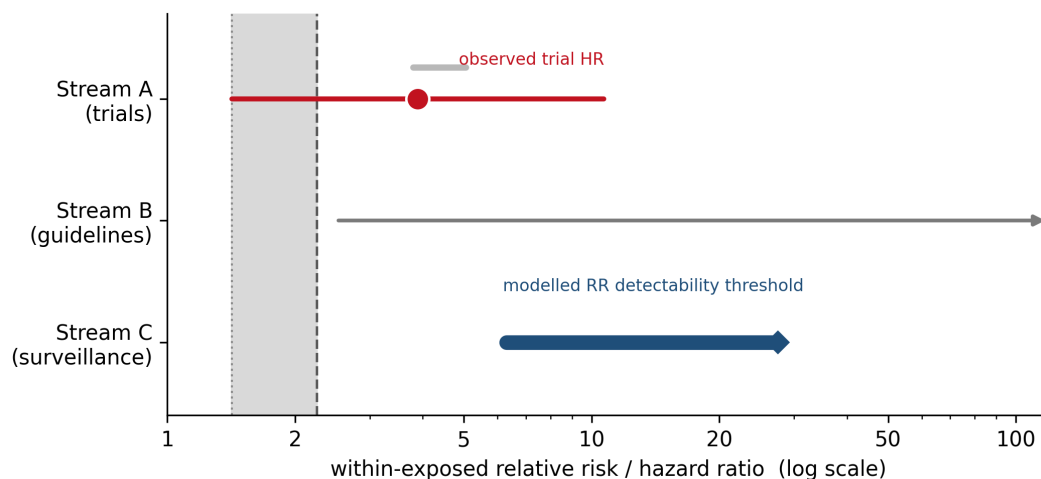


Figure 2: Detectability of the doxy-PEP-associated *S. aureus* resistance signal across the three evidence streams. The shaded band shows the empirical benchmark range for within-exposed relative risk (RR 1.42–2.25). The final DoxyPEP trial reported a hazard ratio (HR) of 3.89 for incident doxycycline-resistant *S. aureus*. Guidance specifies no *S. aureus* monitoring threshold, whereas modelled population surveillance would require an RR of approximately 6–28 under the primary combined PrEP+PLWH exposure specification; the more restrictive PrEP-only specification requires approximately 14–64. The trial HR is shown only as an empirical magnitude reference and is not numerically interchangeable with the modelled RR thresholds because the estimands differ.

to the *S. aureus* tetracycline phenotype at a common denominator (0/12).

The sharper reading is that the trial’s measurement system evolved to meet its own question while the implementation system did not: the same endpoint moved from a cross-sectional fraction read as reassurance to a participant-level HR of 3.89, yet guidance and surveillance did not follow²¹—a demonstrated signal handed to systems built to measure something else.

Tetracycline resistance mechanisms and the phenotype unit

Staphylococcal surveillance is standardised around methicillin, yet the treatment-relevant axis here is tetracycline susceptibility, used empirically before methicillin status is known, so the two axes are orthogonal. The molecular basis of tetracycline resistance is likewise heterogeneous: *tet(K)* efflux leaves doxycycline largely usable whereas *tet(M)* ribosomal protection can compromise it, and the trial resolved incident doxycycline resistance without resolving which.^{3,8} Of 823 invasive USA300 isolates in national surveillance, 72 (9%) were tetracycline-resistant, yet 69 were doxycycline-susceptible *tet(K)* and only 3 doxycycline-resistant *tet(M)*²²—

Table 2: Summary of trial resolution, guidance requirements, and surveillance capability for the doxy-PEP-associated *S. aureus* resistance signal. The final trial resolved an incident doxycycline-resistant *S. aureus* signal, but no assessed governmental guidance required *S. aureus* monitoring and no assessed US surveillance system linked doxy-PEP exposure to *S. aureus* tetracycline phenotype at a common denominator. RR_{needed} is the modelled within-exposed relative risk required to produce a detectable population-level change under the primary surveillance specification.

Stream	Question	Result
A. Trials	Was the bystander signal resolved?	Yes—HR 3.89 (1.42–10.68), incident doxy-R <i>S. aureus</i>
B. Guidance	Is <i>S. aureus</i> monitoring required?	No—0/6 outbreak-matched; 0/10 coded
C. Surveillance	Is exposure linked to phenotype?	No—0/12 systems; $RR_{\text{needed}} \approx 6\text{--}28$ to detect

a single breakpoint that combines mechanisms with opposite implications for doxycycline susceptibility. An inherited surveillance variable can stay valid for its original question while being insufficient for the one now imposed on it; we do not forecast which mechanism is spreading, only that the inherited variable cannot say.

Instrument requirements for population monitoring

Following the signal to the population requires an exposure-linked sentinel instrument the deployed architecture does not provide. Such an instrument would sample *S. aureus* from doxy-PEP-exposed individuals, fixing the exposed fraction and the denominator, and would include the oropharyngeal and peri-anal sites, since nares-only screening under-detects colonisation by about 20% in this population.²³ It would report resistance per carrier and by mechanism (*tet*(M) versus *tet*(K)) rather than as a pooled tetracycline-class rate, and record dosing frequency rather than prescription status. Finally, it would carry *S. aureus* tetracycline susceptibility at a population denominator—most directly by adding it to the platform that already tracks invasive MRSA—joined to exposure on geography, with access to individual-level trial data. These components are technically feasible but are not deployed together at scale.

England’s national doxy-PEP monitoring plan is an instructive out-of-frame case. Outside our prespecified US corpus, it nonetheless proposes the phenotype half we find nowhere—tetracycline resistance in bystander pathogens including *S. aureus*—through routine labora-

tory AMR surveillance.¹³ Yet its specimens come mainly from hospitalised patients; it records
neither doxy-PEP use nor sexual orientation; and its exposure contrast is therefore a male-
to-female resistance ratio in an 18–45 age band, an STI-surveillance conditioning set carried
onto an organism that does not respect it. The same programme already runs exposure-linked
sentinel surveillance that records recent doxy-PEP use—but only for the gonococcus: the in-
strument exists, and inheritance points it at the in-category organism.

These failures share a structure. A guideline or a surveillance system carries in its sam-
pling frame, cadence, coding, denominator and unit the organism and question it was built
for. The doxy-PEP apparatus, built to measure gonococcal efficacy, inherits for the bystander
commensal a denominator the intervention itself moves, a phenotype label spanning two mech-
anisms and a surveillance architecture standardised around methicillin. We call this *mea-
surement inheritance*. A downstream system retains the measurement dimensions appropri-
ate to the upstream question—estimand, denominator, unit, cadence, linkage, measurement
responsibility—but not those the downstream question requires. The question then becomes
unanswerable not for want of data but for want of an inherited dimension. The construct is
built from familiar parts but concerns how they couple *across* stages of an evidence chain.^{24,25}
The downstream gap is therefore not closable by a better analysis of the trial’s data—that anal-
ysis has been done, and it is significant—but is a property of the guidance and surveillance
instruments.

Doxy-PEP’s benefit accrues to the recipient; the tetracycline selection cost falls on people
outside the prescribing decision—outpatients with skin infections and, at the sharp end, dial-
ysis, surgical and oncology patients for whom the loss of an oral anti-staphylococcal option
is consequential. A negative externality that no deployed instrument can price is not thereby
small; it is unweighable. The wider distributive argument—that the two sides are observed on
unequal terms, a property we term *distributive observability*—is developed in the Supplemen-
tary data (S6).²⁶

Limitations

We make no population-level causal claim and no transmission claim: whether the individual-level signal propagates into a population externality is precisely what no deployed system can determine. We did not re-estimate or dispute the trial’s randomised result. Colonisation is not infection, and the colonisation–infection link is itself unmeasured here. Much of the evidence for how *S. aureus* clusters and moves through these networks exists only for its MRSA subset, so where our clustering analysis relies on MRSA we mark it as such. Finally, the trial reaches only a roughly twelve-month horizon; persistence, dose–response and mechanism remain open. Our coding carries two asymmetries. The guidance corpus was double-coded on a 5-document subsample by two passes run through different interfaces—a command-line agent and a chat interface—but the same model family, so their errors are not independent; the 92% concordance measures the reproducibility of the coding protocol, not its accuracy. The surveillance enumeration was coded by a single pass. Consistent with a falsification standard set in advance, the inheritance claim fails if a doxy-PEP guideline comes to require *S. aureus* monitoring, or if a deployed system comes to link doxy-PEP exposure to an *S. aureus* tetracycline phenotype at a common population denominator; a proposed national programme does not yet meet this standard.¹³

Conclusions

The randomised instrument generated a bystander-resistance signal that the guidance and surveillance built around doxy-PEP are not instrumented to follow. The gap is one of measurement architecture, not of missing data, and it is closable—by sentinel surveillance that links doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator. Until it is built, the population question the trial’s authors posed cannot be answered.

Acknowledgements

A large language model was used for writing and analysis-engineering assistance beyond copy-editing. The scientific argument, the reading of primary sources, the model specification and all

conclusions are the author’s own; the author reviewed all generated content, independently ver- 311
ified every numerical value and quotation against primary sources and the regenerable analysis 312
outputs, and takes full responsibility. No artificial-intelligence tool is credited as an author. 313

Funding 314

This work received no specific grant from any funding agency in the public, commercial or 315
not-for-profit sectors. It was conducted as an unfunded investigator-initiated study. 316

Transparency declarations 317

A. C. D. reports former employment at Gilead Sciences (2020–2024) with prior ownership of 318
corporate stock, all divested by 2024; Gilead played no role in this work. A. C. D. is sole owner 319
of Nyx Dynamics, LLC. No other conflicts to declare. All data are public (AIDSVu; published 320
trials and guidelines) and all analysis code is openly available (<https://github.com/Nyx-Dynamics/doxypep-bystander>) and archived at Zenodo (doi:10.5281/zeno 321
do.22747669),²⁷ regenerating every figure and table by a single command; every coded 322
source is SHA-256-pinned. No restricted-access data were used. 323
324

References 325

- [1] Luetkemeyer AF, Donnell D, Dombrowski JC, et al. Postexposure doxycycline to prevent 326
bacterial sexually transmitted infections. *N Engl J Med*. 2023;388:1296-306. 327
- [2] Bachmann LH, Barbee LA, Chan P, Reno H, Workowski KA, Hoover K, et al. CDC 328
clinical guidelines on the use of doxycycline postexposure prophylaxis for bacterial sex- 329
ually transmitted infection prevention, United States, 2024. *MMWR Recomm Rep*. 330
2024;73(2):1-8. 331
- [3] Grossman TH. Tetracycline antibiotics and resistance. *Cold Spring Harb Perspect Med*. 332
2016;6(4):a025387. 333

- [4] Diep BA, Chambers HF, Graber CJ, Szumowski JD, Miller LG, Han LL, et al. Emer- 334
gence of multidrug-resistant, community-associated, methicillin-resistant *Staphylococcus* 335
aureus clone USA300 in men who have sex with men. *Ann Intern Med*. 2008;148(4):249- 336
57. 337
- [5] de Jong GM, van der Boor SC, van Bokhoven C, et al. Methicillin-resistant *Staphylo-* 338
coccus aureus (MRSA) in men having sex with men (MSM): a systematic review. *BMC* 339
Infect Dis. 2025;25:299. 340
- [6] Han LL, McDougal LK, Gorwitz RJ, Mayer KH, Patel JB, Sennott JM, et al. High fre- 341
quencies of clindamycin and tetracycline resistance in methicillin-resistant *Staphylococ-* 342
cus aureus pulsed-field type USA300 isolates collected at a Boston ambulatory health 343
center. *J Clin Microbiol*. 2007;45(4):1350-2. 344
- [7] Stevens DL, Bisno AL, Chambers HF, Dellinger EP, Goldstein EJC, Gorbach SL, et al. 345
Practice guidelines for the diagnosis and management of skin and soft tissue infec- 346
tions: 2014 update by the Infectious Diseases Society of America. *Clin Infect Dis*. 347
2014;59(2):e10-52. 348
- [8] Liu C, Bayer A, Cosgrove SE, Daum RS, Fridkin SK, Gorwitz RJ, et al. Clinical 349
practice guidelines by the Infectious Diseases Society of America for the treatment of 350
methicillin-resistant *Staphylococcus aureus* infections in adults and children. *Clin Infect* 351
Dis. 2011;52(3):e18-55. 352
- [9] Donà MG, Zaccarelli M, Giuliani E, et al. Doxycycline postexposure prophylaxis: a 353
cross-sectional survey on knowledge, attitudes and prescribing practices among Ital- 354
ian dermatologists of sexually transmitted infection centers. *Ital J Dermatol Venereol*. 355
2026;161(2):120-5. 356
- [10] Spence M, Fowler C, Absalom S, Roper T, Williams D, Richardson D. Healthcare pro- 357
fessionals' beliefs and concerns about the use of doxycycline post-exposure prophylaxis 358
(doxyPEP): a systematic review. *Venereology*. 2026;5(1):9. 359

- [11] Mende K, Beckius ML, Zera WC, Yu X, Li P, Tribble DR, et al. Lack of doxycycline antimalarial prophylaxis impact on *Staphylococcus aureus* tetracycline resistance. *Diagn Microbiol Infect Dis*. 2016;86(2):211-20.
- [12] Luetkemeyer AF, Donnell D, Cohen SE, et al. Doxycycline to prevent bacterial sexually transmitted infections in the USA: final results from the DoxyPEP multicentre, open-label, randomised controlled trial and open-label extension. *Lancet Infect Dis*. 2025;25:873-83.
- [13] UK Health Security Agency. Monitoring and evaluation plan of doxycycline post-exposure prophylaxis (doxyPEP). London: UK Health Security Agency; 2026. Guidance; applies to England. Available from: <https://www.gov.uk/government/publications/doxy pep-monitoring-and-evaluation/monitoring-and-evaluation-plan-of-doxycycline-post-exposure-prophylaxis-doxy pep> (accessed 12 September 2026).
- [14] Armond ACV, Cobey KD, Moher D. Research Integrity definitions and challenges. *Journal of Clinical Epidemiology*. 2024;171:111367.
- [15] Soge OO, Thibault CS, Cannon CA, et al. Potential impact of doxycycline post-exposure prophylaxis on tetracycline resistance in *Neisseria gonorrhoeae* and colonization with tetracycline-resistant *Staphylococcus aureus* and Group A *Streptococcus*. *Clin Infect Dis*. 2025;80:1188-96.
- [16] Vanbaelen T, Manoharan-Basil SS, Kenyon C. Studies of postexposure prophylaxis with doxycycline should consider population-level selection for antimicrobial resistance. *Lancet Infect Dis*. 2024;24:e606-7.
- [17] Luetkemeyer AF, DoxyPEP Study Team. DoxyPEP and antimicrobial resistance in *N. gonorrhoeae*, commensal *Neisseria* and *S. aureus*. Seattle, WA; 2023. Conference on Retroviruses and Opportunistic Infections (CROI), Oral Abstract OA-3.
- [18] Mayer KH, Traeger M, Marcus JL. Doxycycline Postexposure Prophylaxis and Sexually Transmitted Infections. *JAMA*. 2023;330(14):1381-2.

- [19] Flores J, Davis AM, Hazra A. Doxycycline Postexposure Prophylaxis to Prevent Bacterial Sexually Transmitted Infection. *JAMA*. 2025;333(3):248-9.
- [20] AIDSVu. State and county PrEP use and PrEP-to-Need Ratio, 2012–2025; 2026. Public data; accessed 2026. Emory University, Rollins School of Public Health. <https://aidsvu.org>.
- [21] Renson A, Montoya L, Goin DE, Díaz I, Ross RK. Pulling Back the Curtain: The Road from Statistical Estimand to Machine-Learning-Based Estimator for Epidemiologists (No Wizard Required). *American Journal of Epidemiology*. 2025;194(12):3566-71.
- [22] McDougal LK, Fosheim GE, Nicholson A, Bulens SN, Limbago BM, Shearer JES, et al. Emergence of resistance among USA300 methicillin-resistant *Staphylococcus aureus* isolates causing invasive disease in the United States. *Antimicrob Agents Chemother*. 2010;54(9):3804-11.
- [23] Kyaw WM, Lee LK, Siong WC, Ping ACL, Ang B, Leo YS. Prevalence of and risk factors for MRSA colonization in HIV-positive outpatients in Singapore. *AIDS Res Ther*. 2012;9:33.
- [24] Tancredi S, Cullati S, Chiolerio A. Surveillance bias in the assessment of the size of COVID-19 epidemic waves: a case study. *Public Health*. 2024;234:98-104.
- [25] Treem JW, Barley WC, Weber MS, Barbour JB. Signaling and Meaning in Organizational Analytics: Coping with Goodhart’s Law in an Era of Digitization and Datafication. *Journal of Computer-Mediated Communication*. 2023;28(4):zmad023.
- [26] Montoya LM, Geng EH, Valancius M, Kosorok MR, Petersen ML. Effects Among the Affected. *Statistics in Medicine*. 2025;44(28-30):e70353.
- [27] Demidont AC. Guidance and surveillance capability for bystander *Staphylococcus aureus* tetracycline resistance under doxy-PEP: a three-stream evidence audit — research compendium. Zenodo; 2026. Software and data compendium.